Additional exercises for simulations
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Problem 1. Consider a simulator that models the arrival and departure of people from the King's Cross plaza. We would like to study (a) the mean number of people who are in the plaza as well as the dynamic evolution of this quantity (b) the mean delay to traverse the plaza and the dynamic evolution of this quantity, and (c) the impact of changing the train schedule on the number of people in the plaza and the plaza traversal delay. What would be the system state variables and counter variables necessary to accomplish this task?

Problem 2. Suppose that the discrete probability distribution over the 5 values $x_i, i \in [1, 5]$, taken by the random variable X, is given by the vector [0.1, 0.3, 0.2, 0.1, 0.3]. Discuss how you could generate values taken on by this variable.

Problem 3. Suppose that the first four calls to a $U[0, 1]$ random number generator return the values, to three digits of precision, of 0.127, 0.632, 0.829, and 0.243. Use these to compute the times of the first five arrivals to a Poisson process with parameter $\lambda = 5.0$.

Problem 4. Suppose you are using simulations to compute the mean number of people on King's Cross plaza and you would like to have 95% confidence that the true mean lies within ± 10 people of the mean value you estimate from your simulations. What stopping rule should you use to decide that you have run enough simulations?

Problem 5. Explain one approach you might use to exploit antithetic variables to reduce the variance in your simulation outputs.

Problem 6. To study the impact of datacenter network capacity on the query latency of a search engine, a researcher builds a simulator using custom code and their own random number generator. However, they notice that the results of this single long simulation run are far from accurate. What might be some reasons for this?